

Hydrogen Sulfide (H₂S) Gas Sensor

0 ppm to 50 – 5000 ppm

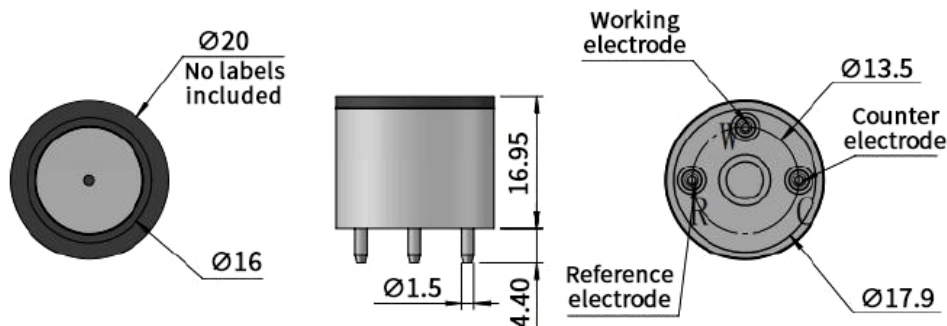
Overview

Select a high-performance H₂S gas sensor, this gas sensor is based on the electrochemical three-electrode principle, featuring 0.1ppm resolution and <30s fast response. The hydrogen sulfide sensor offers flexible detection ranges, supporting selections from 0-50ppm to 0-5000ppm. It exhibits minimal zero drift and excellent long-term stability, ensuring precise readings.

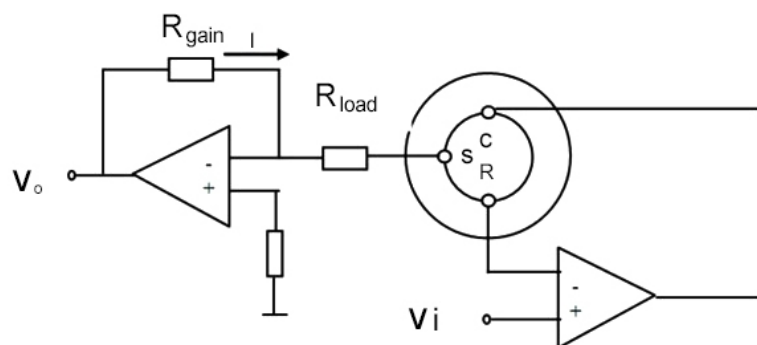
Features

- The hydrogen sulfide sensor offers high precision and extended lifespan, ensuring reliable output during prolonged monitoring.
- The H₂S gas sensor features rapid response and quick zero recovery, promptly reflecting changes in gas concentration.
- Its low-power design combines high sensitivity with a wide linear range, making it suitable for diverse industrial detection needs.
- This hydrogen sulfide gas sensor features strong anti-interference capability, excellent repeatability, and long-term stability. It guarantees precise and stable detection results even in complex environments, making it an ideal choice for safety monitoring.

Dimension (unit: mm)

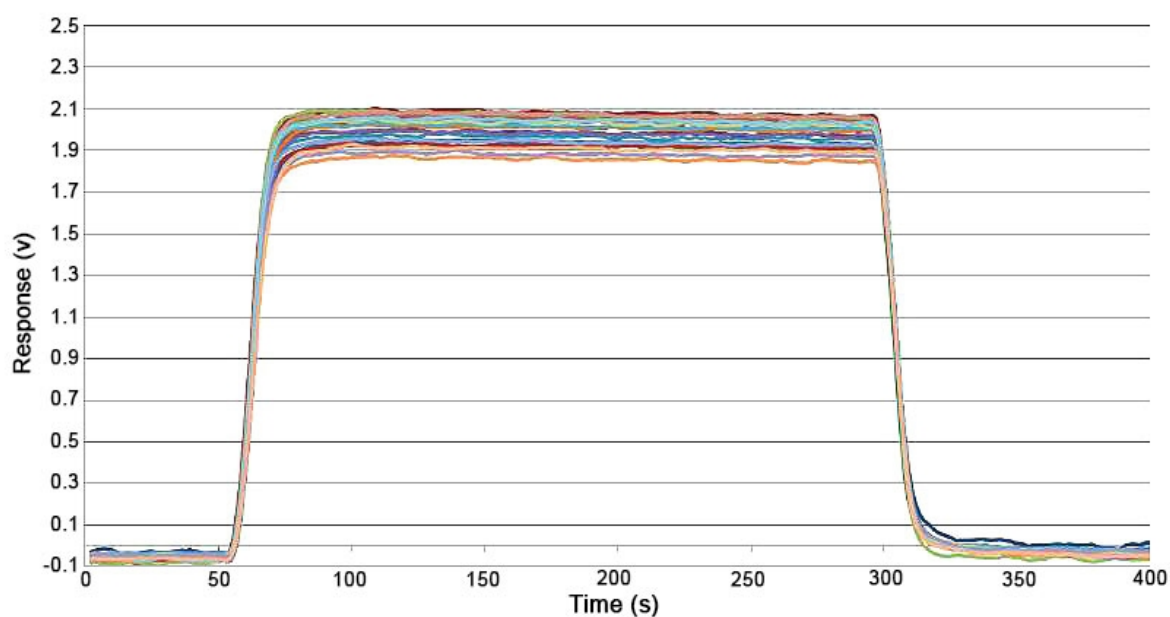


Circuit Diagram



Note: The above diagram shows the basic test circuit for an H₂S sensor.

Response Recovery Curve



Cross Interference

Interface Gas	Gas Concentration Used (ppm)	Displayed Value (ppm H ₂ S)
CO	300	1
SO ₂	20	1
NH ₃	50	0
NO ₂	20	-1
H ₂	1000	0
NO	50	2
Cl ₂	20	-1.4

Specification

Range Select

Detection Range (ppm)	Maximum Load Concentration (ppm)	Sensitivity (nA/ppm)	Zero Drift (ppm)	Resolution (ppm)
0 – 50	100	1200 ± 250	1	0.1
0 – 100	500	650 ± 200	2	1
0 – 500	1000	225 ± 50	2	1
0 – 1000	2000	225 ± 50	2	1
0 – 2000	4000	85 ± 25	2	3
0 – 5000	10000	45 ± 5	2	5

Universal Specs

Detection Principle	Electrochemical three-electrode
Detected Gas	H ₂ S
Detection Range	0-50 ppm, 0-100 ppm, 0-500 ppm, 0-1000 ppm, 0-2000 ppm, 0-5000 ppm
Maximum Load Concentration	100/500/1000/2000/3000/10000 ppm
Sensitivity	1200 ± 250 nA/ppm
Zero Drift	± 1 ppm
Resolution	0.1 ppm
Response Time	< 30 s
Bias Voltage	0 V
Load Resistance	5 – 30 Ω
Temperature Range	-30°C to 50°C
Humidity Range	15 – 90% RH (non-condensing)
Repeatability	1% output signal
Long-Term Stability	< 2% signal/month
Linearity	Linearity, regression coefficient R ² = 0.999
Operating Pressure	90–110 kPa
Service Life	2 years

Application

- Industrial H₂S detection
- Mine H₂S detection
- Atmospheric H₂S detection

Source

<https://www.such.com/hydrogen-sulfide-sensor>